Design for Environment

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Product Design for Environment

- Product Design for the environment is a very contemporary topic.
- Several researchers all over the world are busy attempting to design products that minimize the impact on environment.
- The present chapter is an attempt to motivate young research workers to pursue Green Product Design.

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Introduction

- Product Design for the environment is a design approach for reducing the impact of products on the environment.

- Products can have adverse impact on the environment through polluting processes and the consumption of large quantities of raw materials. & energy

- The impact can be adverse also due to the consumption of large amounts of energy and difficulties during disposal. → plastics
  → one shot product

- Because of this, one must consider a product's entire life cycle, from creation through use through disposal.

- In this life cycle, there are many events of creating pollution and many opportunities for recycling, remanufacturing, reuse, and reducing environmental impact. Reduce, Reuse, Recycle

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Stages of Product Life Cycle

1. To meet this challenge, designers must understand the process of life cycle assessment that adopts a total view by analyzing the entire life cycle of a product, process, package, material, or handling activity.

2. Life cycle stages encompass extraction and processing of raw material manufacturing, transportation, and distribution; use/reuse/maintenance; recycling and composting; and final disposition.

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Importance of DfE

- Design for the environment is an important activity for a design team because environmental damage is greatly influenced in the early design phases.

- 80% of the environmental damage of a product is established after 20% of the design activity is complete. The design for environment is essentially due to three factors:

1. **Customer demand**
2. **Government Pressure**
3. **ISO Requirements**

• Carbon foot print

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Importance of DfE

1. **Customer demand**

- From the business viewpoint, the reason a design team should choose to complete design for the environment is that modern customers are now demanding products with least environmental impact.
- The world's population is becoming aware of society's influence; society is attempting to minimize its impact.
- Creating a product that impacts the environment less becomes a market advantage.

For Example :

Xerox; Xerox has a commitment to be a waste-free company; as apart of its operations and its products, it will generate no waste material that cannot be recycled or remanufactured (Xerox,2000) Xerox aims at 100% recyclable product.

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Importance of DfE

2. Government Pressure

- Government agencies also enforce reduced environmental impact standards for products.
- Such regulatory pressure will only grow with time.
- Many countries now have for products such as packaging computers and transportation vehicles, complete with required recycling of the components.

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Importance of DfE

3. ISO Requirements

- Standards are also being developed to support design for the environment as a practice.

- Underlying all of these activities are the market forces that demand design for the environment as a necessary part of modern product development.

MNC → paperless → E - governance.

Product - information

certification body

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Environmental Factors

- It is important to understand pollution types, their range of impact, and what can be done in product development to reduce the impact of pollution.

- There are many guidelines developed to help industry understand and deal with its impact on the environment.

- Coalition for Environmentally Responsible Economies (CERES), as the CERES Principles establishes objectives and guidelines to help companies maintain good environmental performance.

- These principles were adopted by the Exxon Corporation (March 24, 1989) and later adopted by hundreds of companies.

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Outline of the Objectives :

- Companies will reduce the release of pollutants that endanger the earth.
- Consistency of use of resources: Companies will use raw materials at a level where they can be sustained.
- Companies will minimize waste wherever possible. When waste cannot be avoided, recycling will be adopted. → waste water are recycled.
- Companies will use environmentally safe energy and invest in energy conservation. → wind/ solar / tidal
- Companies will minimize health risk to employees and the community.

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Outline of the Objectives :

- Companies will sell products that minimize environmental impact and are safe for consumers to use.

- Companies will take responsibility through cleanup and compensation for environmental harm.

- Companies will disclose to employees and the community incidents that cause environmental harm or pose health and safety hazards.

  ↳ iron & steel
  ↳ Cu extra

- At least one member of a company's board will be qualified to represent environmental interests, and a senior executive for environmental affairs will be appointed.

  Board → member → Environmental Representation.

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Outline of the Objectives :

- Companies will conduct annual self-evaluations of progress in implementing these principles and make results of independent environmental audits available to the public.

- Pollution can be organized by the scope of their environmental impact, from global to regional to local impacts.

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Scope of Environmental Impact

1. Global Issues

- There are pollution problems which exist on a global scale. These include concern over climate change, ozone depletion, and biodiversity loss.

- The concern over climate change is because of the probable consequences of possible large changes in the earth's climate due to increase in greenhouse gases.

- This is due to burning of fossil fuels which increase carbon dioxide levels in the atmosphere. From the product design point of view, developing products that use less energy will help mitigate this problem.

IC Engine → Electric Car → CO_2

→ Energy density

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Scope of Environmental Impact

1. **Global issues**

- Another global pollution concern is the depletion of the ozone layer. The ozone layer is a thin layer in the upper atmosphere that blocks ultraviolet radiation from reaching the Earth's surface.

- Fluorocarbon gases, from our industrial society, may also react with and reduce the ozone gas in this layer. From a product design viewpoint, developing products that do not make use of or release these harmful chemicals, either in use, manufacture, or disposal, will help to solve this problem.

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Scope of Environmental Impact

2. Regional and local issues

- Other environmental problems exist on a more regional level. These include problems of acid rain, where pollution by products in one region can cause acid rain in another region.

- Air pollution and smog also are regional problems. Water pollution, either in the ground water, river, bay, or ocean, is also a regional problem, often caused by herbicides and pesticides, in addition to suburban and urban street water run-off.

- Other contaminants can enter through streams and landfills as water pollution. Herbicides and pesticides are typical problem compounds whose amounts introduced to a regional area must be controlled.

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Three R's (Reduce, Reuse, Recycle), Renewable energy

- It's the mantra recognizable by its distinct the arrows in the triangular pattern.

https://www.suntory.com/csr/activity/environment/reduce/resources/recycle/

Three R's (Reduce, Reuse, Recycle)

Reduce your waste

Domestic front

- Closely monitor your inventory supply and look for ways to improve your process.

- Research is key!
  - Can you switch to a water based, or less toxic ingredient?
  - Can you switch to synthetic oil and extend the time between oil changes just a bit?

- Buy only what you need so that you do not have to worry about expired unused chemicals. → medicine

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Three R's (Reduce, Reuse, Recycle)

http://rethinktires.ca/5-eco-friendly-tips-to-follow-to-reduce-waste/#sthash.QOkBLWl6.dpbs

Three R's (Reduce, Reuse, Recycle)

Reuse what you can

- Again, do your research!
  - I read that you can extend the life of buffered hydrogen fluoride (BHF) in semiconductor manufacturing with ammonia and water evaporated from the BHF.
  - The etch rate can be maintained and the BHF is reused when it was previously only used once per application. Do your research, there is information out there that pertains to you.

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Three R's (Reduce, Reuse, Recycle)

Recycle is the golden child of the three

- There is recycling in the traditional sense, where you might send your spent oil for fuels blending and it is recycled as a fuel.
- Or perhaps you send your fluorescent light ballasts/electronics to a facility where it is dismantled and the parts are recycled separately.
- However, there are other wastes, that are not hazardous wastes, that also have the ability to be recycled.
- I have a client who sells the crushed glass from flawed products on their line to another consumer who makes something new out of their waste.

http://www.leegov.com/solidwaste/residential/recycling

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Three R's (Reduce, Reuse, Recycle)

Recycle is the golden child of the three

- For example,

  - One three-shift soft drink bottling plant can easily generate 20 million gallons of line lubrication wastewater per year. However, there are technologies that enable substantial gains in sustainability by reducing water for lubrication from hundreds of gallons per shift down to mere ounces.

  - A leading producer of biscuits, cookies and crackers implemented a technology that allowed for less adhesive in sealing packages for multiple lines of cookies and biscuits, reducing adhesive consumption by 70%. As result, the company saved 35 barrels of oil, a significant amount of energy to run the line, and more than 260,000 gallons of water previously used for traditional package glue.

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Life Cycle Analysis

- Regardless of the name, the primary aim of life cycle analysis is to identify the environmental impact of the materials and resources used in the manufacture and use of a product.

- This is similar to a procedure that financial managers call sources and uses . Large publicly traded companies will include a "sources and uses of funds" statement in their annual reports.

- The resource in this case is money — where it is obtained, its source, and how it is used to carry out the activities of the business.

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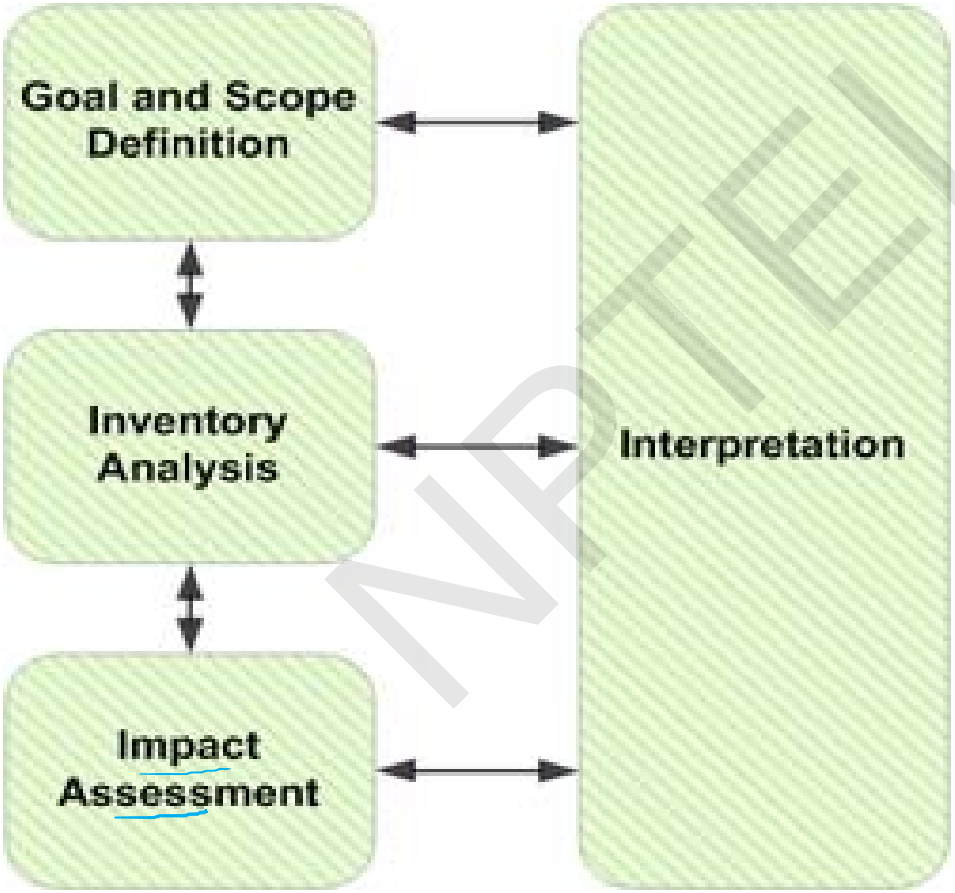
Life Cycle Analysis



https://en.wikipedia.org/wiki/Life-cycle_assessment

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Life Cycle Analysis

- Individuals and institutions that are contemplating lending money to a startup company look for a sources - and – uses worksheet because it is an excellent summary of the 'new venture' financial plan. In a similar manner an LCA can be viewed as a sources - and - uses statement.

- Most LCAs include a comprehensive listing of the inputs, the resources.

- The output defines how effective the facility is in converting these resources into products while minimizing waste.

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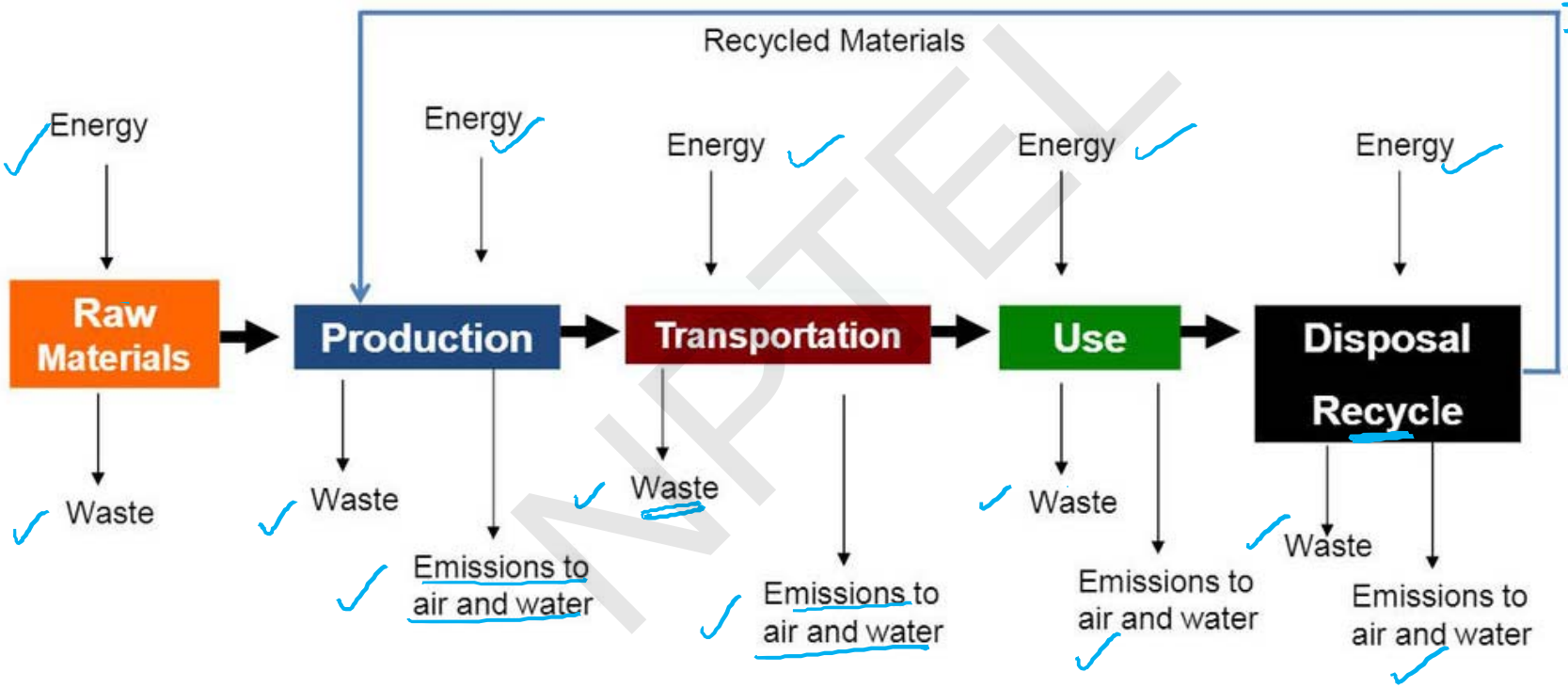
Life Cycle Analysis

1300 kgs

450 students
→ 100 kgs of bio man waste for 3 session.
1 session => 45 kg
→ 45000/450 100 gms
50 gms.

Example LCA Process



http://www.franciscoblaha.info/blog/2017/3/10/the-use-of-life-cycle-assessment-lca-in-fisheries

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Life Cycle Analysis

- Inputs include all raw materials, stocks, and resources that are used for the creation of the product.

- Resources include energy demands (electricity, gas, oil, coal, etc.) and water. In some special instances land use might be included.

crises

- While land is not considered a consumable in the creation of a stock or product, there could be a circumstance that would make the land unusable for a period of time. → Pesticides/fertilizers in agri cultural.

For Example : Strip Mining

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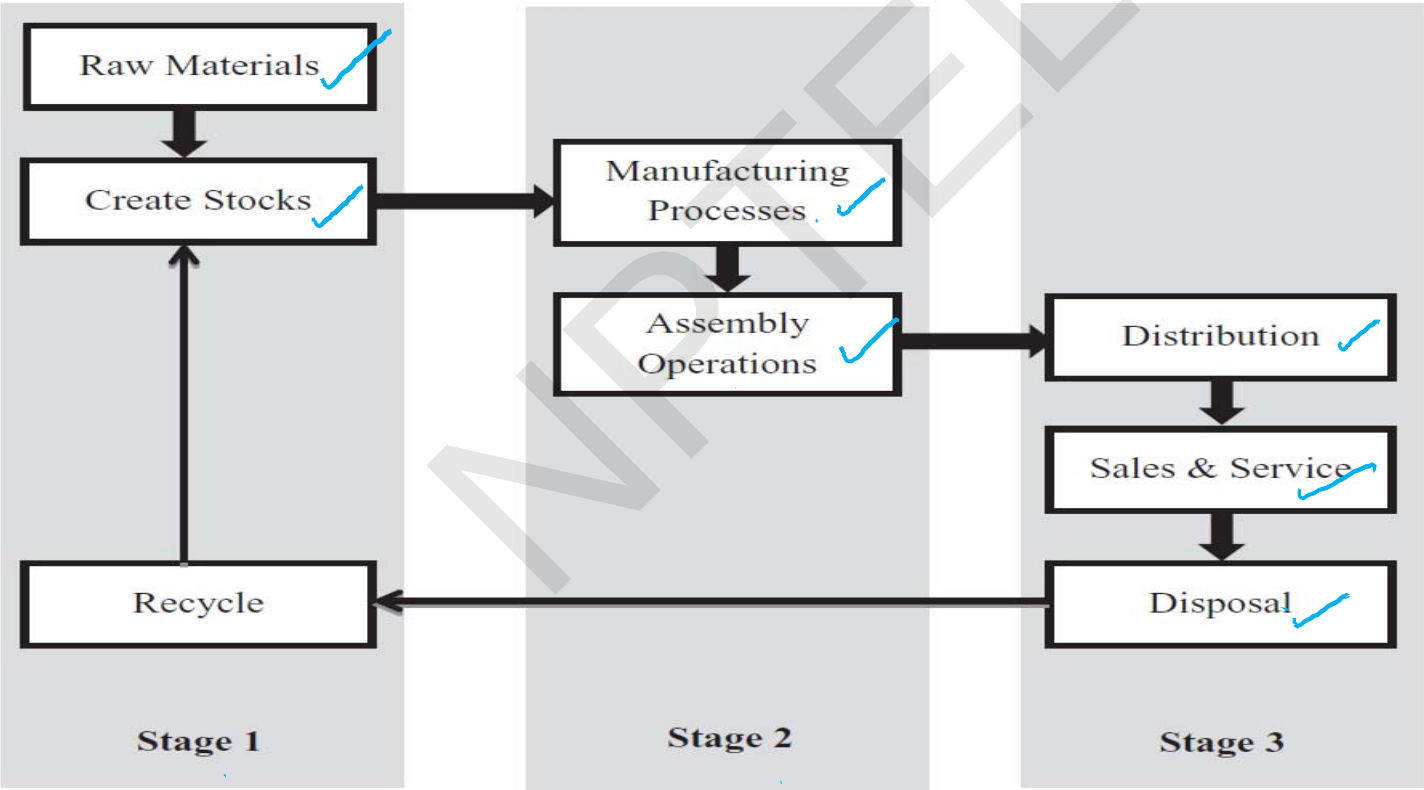
Life Cycle Analysis

Manufacturing { Part, Assembly

- The sequence of manufacturing



Source :Improving Profitability Through Green Manufacturing By David R. Hillis And J. Barry Duvall.

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Life Cycle Analysis

Packing — Primary → Secondary → Tertiary

- This format includes the most common steps in the manufacturing sequence plus extraction of raw materials and repair or service.
- The outputs of course include the product as well as everything else, which is defined as waste. The major waste categories are water and water effluents; airborne emissions; solid waste; and recyclables.
- An item that is often overlooked in the analysis of manufacturing waste is packaging. This is not the case in LCA.
- A major source of waste in the distribution step of an LCA is single-use packaging.
- The LCA, like the manufacturing sequence, addresses material in the first two steps. On the left side of above Figure these steps are identified as Stage 1.

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Life Cycle Analysis

- In the first two steps of an LCA (extraction of raw materials; creation of the stock) the industry carries the name of the material being converted to a stock.

  For Example :

  when someone says " the steel industry " what comes to mind? In most cases an image of a steel mill will pop up in our mind 's eye.

- However, when it becomes a coil of cold-rolled steel it is a stock that will be used to manufacture a product.

- So, beginning with the third step (Stage 2 in above figure) of the LCA, the industry name changes from the material name to the product name.

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Life Cycle Analysis

- Steel would be replaced by a product name such as auto or appliance.
- It is apparent that completing an LCA on just one group of materials or a single industry such as the appliance industry is a major undertaking involving hundreds of companies.

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Life Cycle Analysis

- However, it is the approach that the analysis uses that is valuable. Measuring and quantifying the costs of all the materials and resources required to create a product is a basic part of manufacturing.

- Cost accountants have been allocating direct and indirect costs to specific products and work centers for more than a century.

- An example of a direct cost is the amount of a stock used to create a product. These direct costs and their proper allocation to a product are relatively easy to calculate.

- Indirect costs are more difficult to assign. They are expenditures that are not apparent by examining the bill for materials or the list of operations used to make the product.
- The classic example of an indirect cost is the person who sweeps the aisles when a shift is over.

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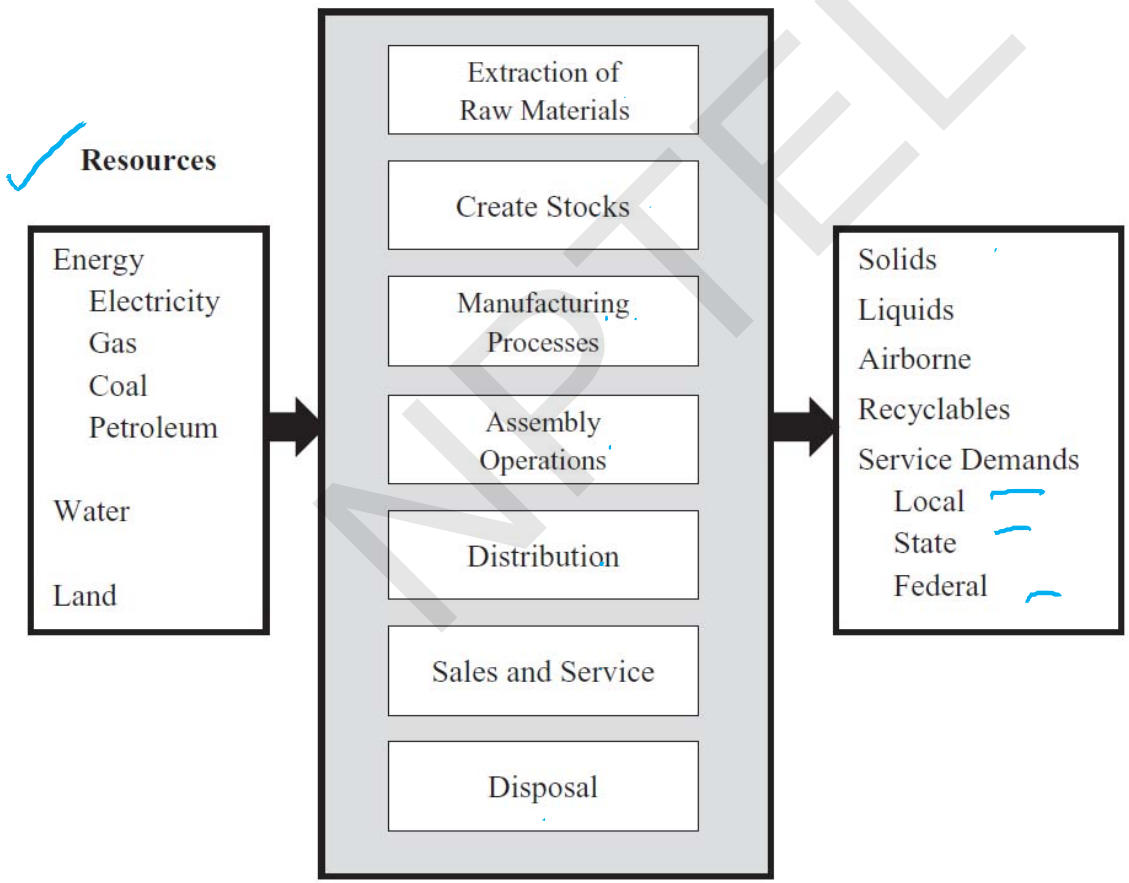
Life Cycle Analysis

A Life cycle analysis model



Source :Improving Profitability Through Green Manufacturing By David R. Hillis And J. Barry Duvall.

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Life Cycle Analysis

- The general approach of the LCA would be useful for building a new plant.
- It is not difficult to list the activities at each of the seven steps of a manufacturing sequence for constructing a manufacturing plant.
- You can list the stocks and processes used to construct the building and the assembly operations to put in the electrical distribution system, the HVAC, the plumbing, and so on that are needed to complete the facility.
- Servicing the building and its final disposal can also be handled effectively.

- **Therefore, the LCA is an excellent technique for assisting management in costing and designing an environmentally effective manufacturing facility**.

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Potential for Waste and Value Added

- Each of the seven major activities in the manufacturing sequence offers manufacturers opportunities for creating value and waste.
- Table below lists these opportunities along with their potential for creating waste. This potential will vary for each of the seven steps.
- For instance, an assembly operation may generate some waste but generally the environmental impact will be minimal.
- However, in some of the other steps the waste and environmental costs can be quite high. As an example for the extraction of raw materials a large part of all waste will be environmental costs.

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Potential for Waste and Value Added

Potential for creating waste compared with the value-added potential

Manufacturing Sequence	Potential for Creating Waste	Potential for Adding Value
Extraction of raw materials	High	Moderate
Create stocks	Moderate to high	Moderate
Manufacturing processes	Moderate to High	High
Assembly operations	Moderate	Low to moderate
Distribution	Low	Low
Sales and service	Low	Moderate
Disposal	High	Low

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Source :Improving Profitability Through Green Manufacturing By David R. Hillis And J. Barry Duvall.

Potential for Waste and Value Added

True or not, he eventually concluded that effective large-volume manufacturing has four principles:

1. The product uses interchangeable parts; no custom fitting or modifications should be required.

2. The product moves to each workstation at a predetermined rate; this was the introduction of continuous flow manufacturing.

3. The work to manufacture the product should be broken into a sequence of simple easy-to-learn tasks. → automate

4. Reducing or eliminating waste of all kinds is an ongoing effort.

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Waste and its unexpected sources

- Too often companies feel that the most effective way to differentiate their product and maintain sales levels is through price, specifically price reduction.

- After the price is reduced the company then looks for ways to reduce its manufacturing costs so that it can remain or once again become profitable.

- Too often organizations try to reduce cost by taking away value from the product. That is wrong.
- **Reducing value by eliminating features, service life, or functionality is placing the burden of cost reduction on the customer when it should be the organization's responsibility.**

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Waste and its unexpected sources

The First Source of Waste

- The first major source of waste originates in the way the company makes its products.
- Much waste is due to the product design and the manufacturing processes used in the plant.
- Certainly the type of materials a company uses, which is a function of the product design, will dictate the plant design and the processes.

Chair
metal
wood
plastic

- Each material and the associated manufacturing processes have their own set of waste parameters that defines the facility.
- But some of the waste occurs due to the organization of the facility and the operation norms that have been established.

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Waste and its unexpected sources

The First Source of Waste

The following are examples of items that would contribute to this source of waste:

Renewable → Roof top energy

Energy meter ↔ Bi-directional

- ***Resources*** . The fuels needed to operate the processes machines, and equipment used in the manufacturing sequence. This would also include plant and office heating, lighting, and air conditioning.

- ***Water*** . This also includes the associated costs of sanitary and storm sewer services.

Chair → plastic → no of process

no of m/c

no of operators

no of sanitary/toilets

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Waste and its unexpected sources

The First Source of Waste

- ***Supplies*** . The secondary materials that are required to complete a manufacturing operation or process but do not become part of the product. An example might be cutting tool coolant/lubricant, towels, cleaners, copy paper, etcetera. These are waste materials that are " accepted " as being part of the manufacturing operation or process.

- ***Wages paid*** . Payment to individuals and to contractors or suppliers who do not add value to the product. This is one of the most difficult categories to control.

injection mould → low cycle time
- maintaina free
- consistence the product
- Quality high.

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Waste and its unexpected sources

The Second Source of Waste

- The second source of waste comes directly from manufacturing operations.
- This source has received the most attention since factories were established to produce products.

- Over the past few decades the people involved in controlling this source of waste now include manufacturing operations and engineering; human resources; and training and development.

- There are a lot of names and acronyms for programs, techniques, and tactics used to reduce waste in manufacturing.

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Waste and its unexpected sources

The Second Source of Waste

by decade

- The concept of lean manufacturing has become the most comprehensive approach for waste reduction now being employed.

- In general a lean manufacturing program works to reduce eight types of waste: A starting point for reducing waste is to conduct a " check - up " not unlike the way a physician conducts an annual physical.

- There are tests to be conducted and information to be gathered along with a physical examination.

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Waste and its unexpected sources

The Second Source of Waste

- The manufacturing facility has to be examined in a similar manner. A helpful approach is to use a checklist for developing a diagnosis — an assessment of the problems from the eight sources of waste:

1. ***Waste from Overproduction*** : More products are produced than required by the customer.

2. ***Waste from Transportation*** . There may be excessive movement of the product or its components during the production process.

o Try to use resources which are available with you while developing/manufacturing products

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Waste and its unexpected sources

The Second Source of Waste

2. ***Waste of Motion*** . Waste of motion occurs when the operator has to look for tools or information, to make adjustments o repairs, to free jams, or to fill out incentive tickets or routing sheets.

4. ***Waiting*** . Time may be wasted, for example, waiting for setup to be completed, materials to arrive, or equipment to be repaired.

5. ***Work-in-process*** . Work - in - process (WIP) includes all stocks, components, and subassemblies in the manufacturing system.

100 → 1 part → B/ import
99 → assembly

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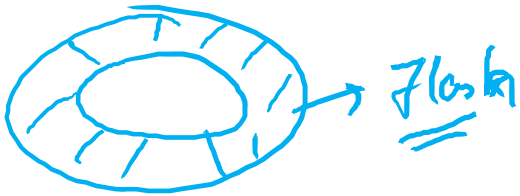
Waste and its unexpected sources

The Second Source of Waste

6. ***Handling*** . After manufacturing the handling of the finished product is done and it also becomes the source of waste when it takes more space and time.

7. ***Defects*** . This includes spoiled or rejected parts, subassemblies, finished goods, returns, warranty work, and product recalls.

8. ***Scrap*** . Material stocks are turned into scrap because of the product design or manufacturing process. Examples include paint overspray, punchings, trimmings, and end-of-reel fragments.



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Waste and its unexpected sources

The Third Source of Waste

- The third major source of waste is the materials and activities that are part of the value - added operations and processes used to manufacture the product.

- To minimize this source of waste requires the designers of the product and the manufacturing engineers and technicians to optimize the materials and processes needed to produce a product.

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Waste and its unexpected sources

The Third Source of Waste

- This source of waste is controlled by three groups of people.
    1. The first group specifies the product's characteristics and function .
    2. The second group uses this specification to design the product and works with
    3. The third group to specify how it will be manufactured.

- These three groups (functions) of people involved in this activity are the following: marketing; product design and engineering; and manufacturing engineering and operations.

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Waste and its unexpected sources

The Third Source of Waste

- They are the ones who will select the materials and processes used to manufacture the product.
- A basic precept in design states that the selection of a material defines the manufacturing processes that will convert the material stocks into a product.
- Therefore for a new product the reduction of waste and emissions begins with material selection.
- Of course someone will be quick to point out that there are several manufacturing processes that can perform a specific operation for a given material.

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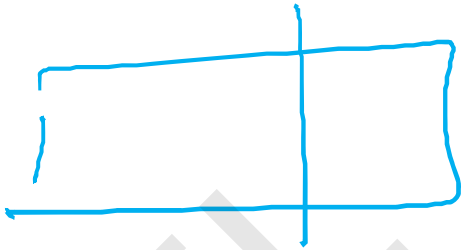
Waste and its unexpected sources

The Third Source of Waste



- For example, suppose the product design requires that a metal plate must be cut in half. The basic process is called *separating* .

- There are several ways to carry out this process but we'll consider just three ways; sawing, shearing, and flame cutting. → smoke → quality of surface is poor

- Of these three, shearing would probably result in virtually no material waste, minimal energy use, and no primary emissions.

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Waste and its unexpected sources

A new product, first phase for waste reduction

- For a new product the first phase of waste reduction involves marketing, product engineering, and manufacturing engineering to make these decisions.

- The below figure shows the groups involved and the inputs guiding their decision making. These inputs, some might call them constraints, form the criteria that shape the design.

- How well each group responds to these inputs in defining the product, materials, and processes will determine the potential for waste and its associated costs.

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Waste and its unexpected sources

A new product, first phase for waste reduction

- Consequently it is critical that they recognize the sources of waste and examine alternative designs that meet the customer's requirements.

- The objective is to create a design that can be produced profitably and be environmentally compliant.

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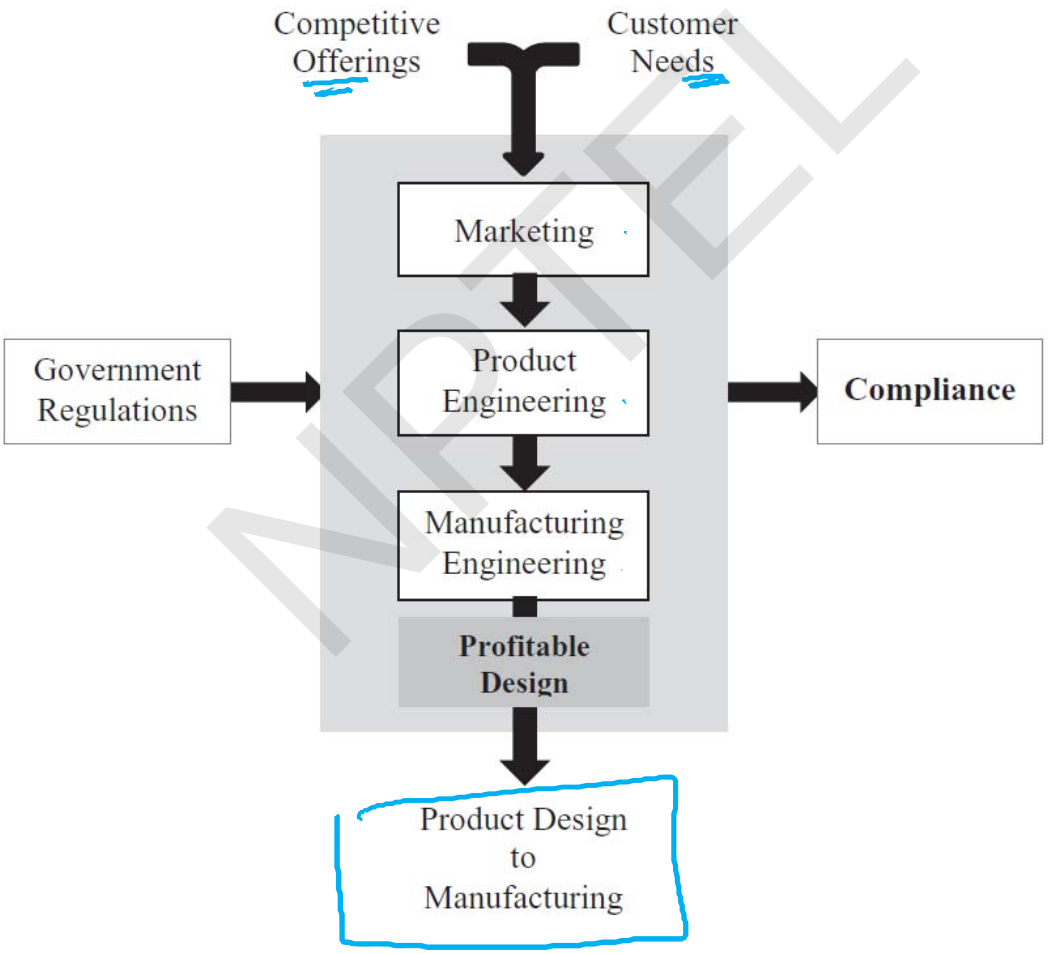
Waste Reduction

The functions and inputs responsible for the design and production of a profitable and compliant product.



Source :Improving Profitability Through Green Manufacturing By David R. Hillis And J. Barry Duvall.

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Waste Reduction

A new product, first phase for waste reduction

- A manufacturer that has a product design function (note that many manufacturers do not design the products they make) has to maintain an engineering database that reflects the current state-of-the-art in materials and process technology.

- The point being that government regulations are continually impacting material stocks and the way they can be processed or used.

For example:

lead - based paints are no longer included in the engineering database for a furniture manufacturer.

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Waste Reduction

Existing products, second phase for waste reduction

- The organizational functions taking the lead in developing the strategy will change from design engineering to production and manufacturing engineering and from marketing to human resources and training.

- Below figure shows the groups responsible for this stage of manufacturing, producing an established design profitably and in compliance.

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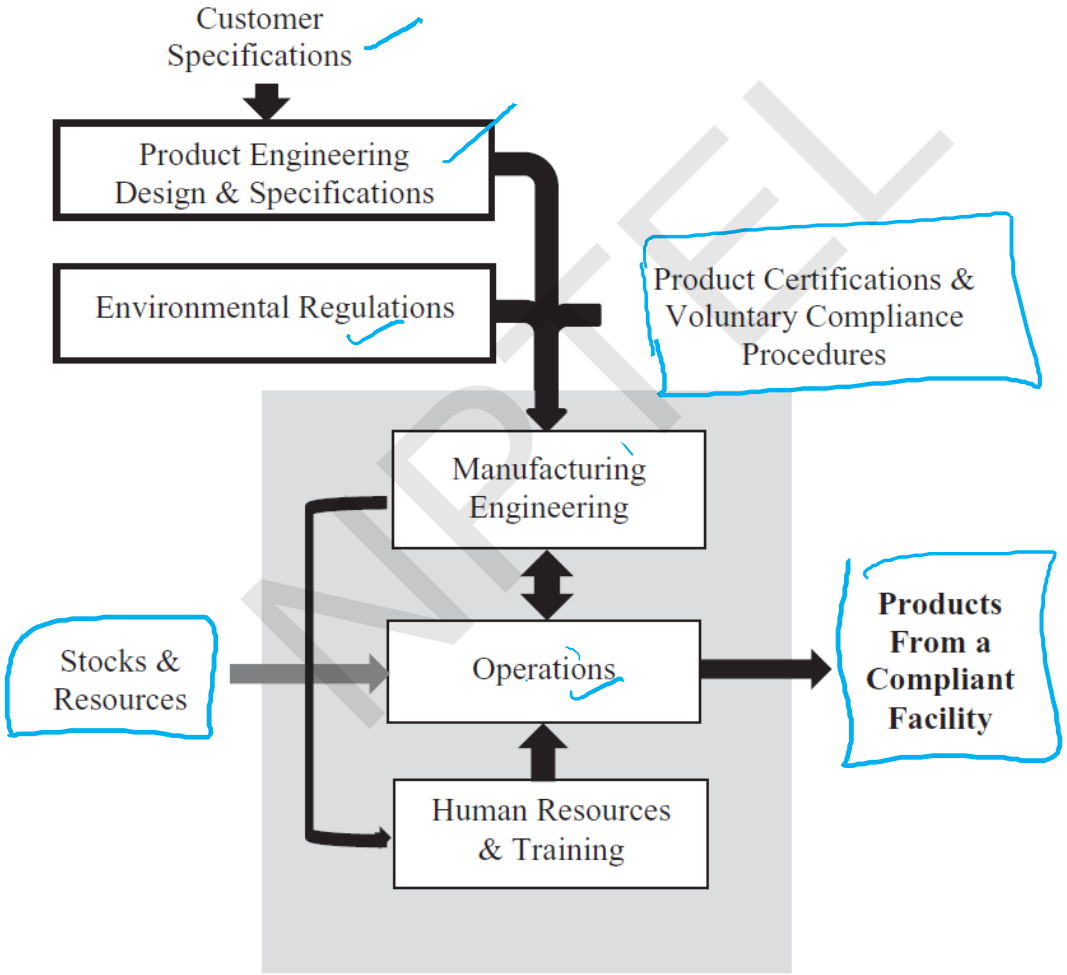
Waste Reduction

The groups involved in reducing waste in an established product design.



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Waste Reduction

Existing products, second phase for waste reduction

- These are the people that will take on the responsibility for reducing waste and ensuring that the manufacturing facility is in compliance.
- If you think about it, this second phase of waste reduction seems to capture more attention because more often product designs tend to be taken as a given.
- This is true in part because underwriter certifications, life testing, and other forms of product performance qualifications make it more difficult to obtain significant changes in an existing product design.
- However, there is more latitude in changing processes and operations than there is in changing a stock.

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Waste Reduction

Existing products, second phase for waste reduction

- But these changes should be done in coordination with the product design group.
- Nonetheless there are plenty of opportunities for waste reduction at this second stage.
- The point is the strategy for environmental compliance should not be limited to abatement, permitting, and lawful waste disposal.

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Regeneration

- Organizational fat can be detected by observing how an organization handles problems. When a major problem arises does management handle it by creating a new department or group to deal with it?

- If this is management's approach it makes the problem an ongoing fixed cost causing the organization to swell in size with these specialty groups.

- Problem solving is a basic function of line management not specialty staffs. Management must clear away its organizational structure as it should clear away out-of-date equipment and stocks.

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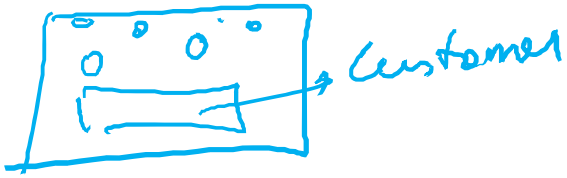
Regeneration



- Ideally a plant should be operated in the same way as a convention center or theater for the performing arts.
- That means it is set up for the event that is currently running but can be quickly changed over to handle a new event that is totally different.
- Each event will (it must) use profitable and environmentally compliant manufacturing processes. Unfortunately most plants are fixed hard - wired facilities.
- These facilities have so much inertia that it is nearly impossible to make meaningful improvements quickly or efficiently.

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Regeneration

- One of the first objections from managers when asked why they are not adopting a more effective method for producing their product is cost.
- They explain the 'price tag' for new equipment and processes makes it prohibitive and they can't afford or don't have the money to invest. This argument misses the point.
- First of all buying new equipment and trying to squeeze it into some corner of a packed manufacturing floor is not the solution.
- The first step is not buying new equipment but getting rid of the waste in the system.

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Regeneration

- comprehensive waste reduction program makes money that later on can be used to update processes or purchase new equipment and locate it where it should be, not where it can be squeezed in.
- One starting point for regeneration is adopting the concepts and techniques that are known as lean manufacturing.
- Each of the three sources of waste provides a starting point and the reasons for adopting the values of lean manufacturing

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Life Cycle of Manufacturing Facility

- Life cycle is actually another term for a life span. If you drive around the outskirts of any large city in the Midwest or Northeast one can see the sad sight of huge crumbling factories.

- But it 's not limited to just these areas of the country; during every economic downturn there are factory buildings old and new that go vacant.

- It is part of the life cycle of manufacturing. Unless an organization is committed to regeneration it will cease to exist.

- Buildings like people have a finite working life. Companies and organizations are no different unless they become adept at regenerating themselves.

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Life Cycle of Manufacturing Facility

- One plant manager explained that a company should never own a building because it is a constraint to its ability to regenerate itself.

For Example :

An apt analogy was the experience of a couple that owned a house that they had lived in for 20 years. An excellent business opportunity became available for them in a town 45 miles away. They took advantage of the opportunity but decided to commute each day instead of moving to the new location. So putting social and family ties aside, the transportation costs and the time spent commuting to avoid relocating became waste caused by the inertia of property ownership.

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Implementing a DfE Process

- The environmental reviews will require design teams to review the checklist of key requirements and to consider guidelines for reducing environmental impact.

- When the DfE process is first implemented, design teams will have to fill out the environmental scorecard after the product design is complete.

- Doing this begins the process of recording environmental data and allows design teams to adapt gradually to the new process.

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Implementing a DfE Process

- The lessons-learned summary will provide the innovation statement metric.
- Below figure shows the Safety Review Process and Environmental Review Process running in parallel.
- The following sections discuss the aforementioned environmental activities in more detail.
- Note that, throughout this process, many other product-development activities are occurring, causing changes to the product design.

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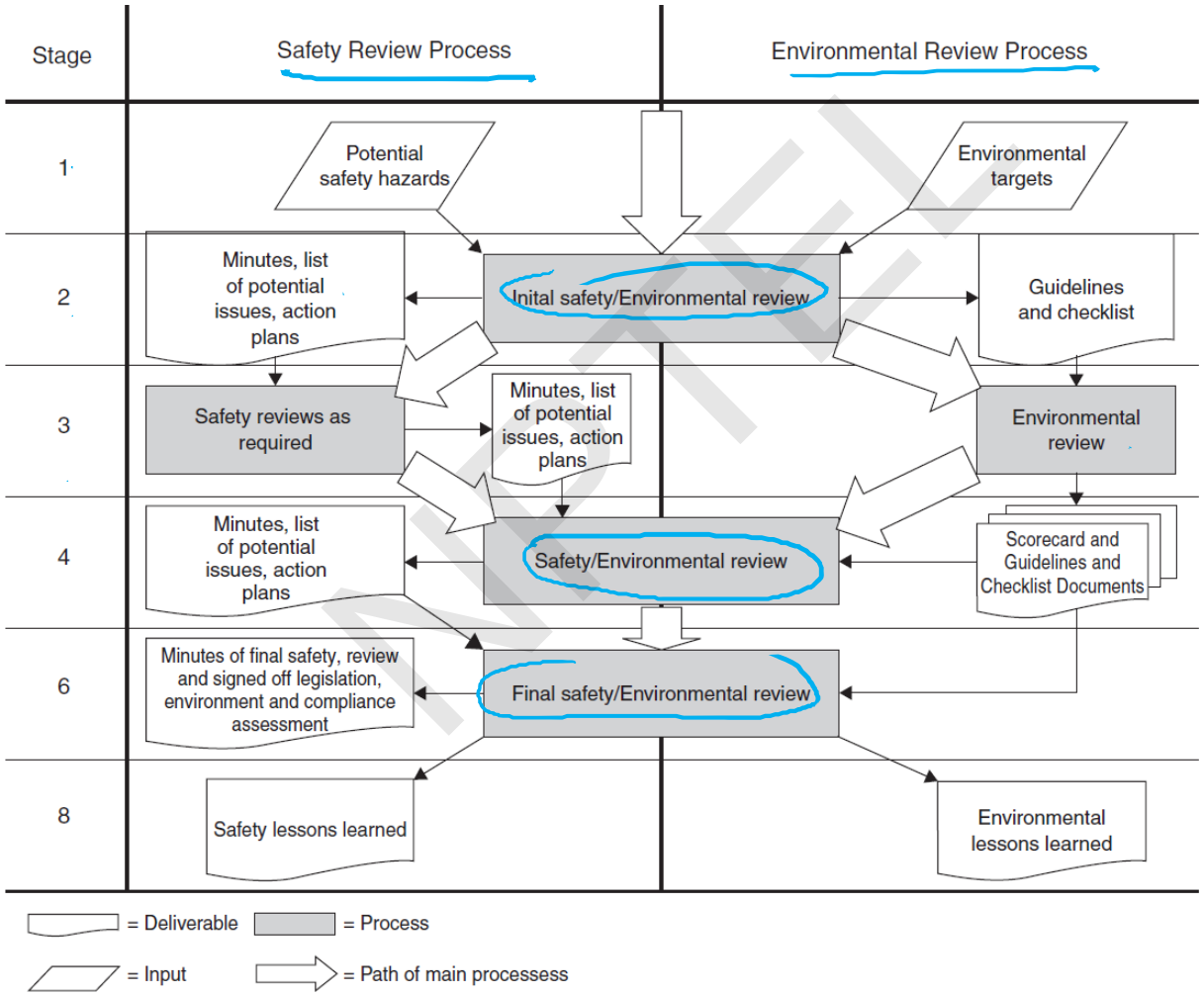
Implementing a DfE Process

Combined safety and environmental review process.



Source : Design For Environment (DfE): Strategies, Practices, Guidelines, Methods, And Tools By Daniel P. Fitzgerald

Product initiation document

- The product initiation document is a document that Black & Decker uses to benchmark competitors, define performance targets, and predict profitability and market share.

- Product initiation document will also address environmental regulations and trends and opportunities to create environmental advantage.

- Targets for environmental improvement will also be included.

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Initial environmental review

- **The first environmental review** is coupled with a safety review.
- During this meeting, the design team should discuss current environmental regulations, design guidelines, and environmental metrics.
- A list of regulations and design guidelines can be found in the guidelines and checklist document.

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Initial environmental review

- The environmental metrics are located in the environmental scorecard. Old lessons learned documents from similar products will be reviewed during this meeting to facilitate environmental design ideas.

- The result of the meeting is an initial assessment plan that includes the tests to be conducted and the analysis to be performed. The reliability representative will write the assessment plan.

- A list of brainstormed ideas for environmental improvement and any other minutes will be included in the assessment plan.

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Conceptual design environmental review

- **The second environmental review** is held separately from the safety hazard review.

- During this meeting, the project team will check compliance regulations, fill in the guidelines and checklist document, discuss the metrics in the scorecard, and review opportunities and additional environmental issues.

- The result of this meeting is an updated guidelines and checklist document and meeting minutes. The reliability representative will update the guidelines and checklist document and write the minutes.

- The lead engineer will update the scorecard for the next meeting.

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Detailed design environmental review

- **The third environmental review** is coupled with a safety review. During this meeting, the project team should ensure that all environmental compliance issues are resolved.
- There should be no further changes to the design due to environmental reasons after this meeting.
- The result of the meeting is an updated guidelines and checklist document and meeting minutes.
- The reliability representative will update the guidelines and checklist document and write the minutes.
- The lead engineer will update the scorecard for the next meeting.

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Final environmental review

- **The fourth and final environmental review** is coupled with a safety review.
- During this meeting, all environmental compliance issues must be resolved.
- Optimally, no design changes due to environmental reasons would have been made between the last meeting and this meeting. The result of the meeting is a final guidelines and checklist document and meeting minutes.
- The reliability representative will finalize the guidelines and checklist document and write the minutes.
- The lead engineer will finalize the scorecard and create a Material Declaration Statement (MDS) packet for the product.

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Postlaunch review

- Include a lessons-learned summary in their product development process.

- This document discusses
  - what went well with the project,
  - what didn't go well with the project, and
  - reasons why the product didn't meet targets set in the trigger document.

- The lessons-learned summary will include environmental design innovations realized during the product development process for publicity and customer questionnaires.

- An example of an item to be included in the lessons learned summary is a materials selection decision.

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Feedback loop

- The completed guidelines and checklist documents and lessons-learned summaries create a feedback loop for the DfE process.
- Design engineers working on similar products can use this information to make better decisions immediately.
- The information is also valuable when the next generation of the product is designed years down the road.
- Design engineers will record what environmental decisions were made and why they were made.

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Feedback loop

- The decision information, scorecards, and comments on the guideline document will be archived permanently.

- The goal is to save the right things so the information is there in the future when more feedback activities, such as a product tear-down to verify scorecard metrics, can be introduced.

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To recapitulate:

1. Why product Design for environment is necessary?
2. What are the stages and importance of design for environment?
3. Which issues comes under the scope of environment?
4. What is the manufacturing sequence in life cycle analysis?
5. What are the main three sources of waste?
6. How to implement a DfE process?
7. What is post-launch review, and feedback loop for DfE?

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Task for students

- Try to take Carton box used in Computer/desktop - CPU and make IS products which can be sold. The products need/not be the same. Try to record customer feed back for further revision in design.

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Thanks

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Lecture 21

Quality Control

Dr. Janakarajan Ramkumar
Professor
Department of Mechanical & Design Program
IIT Kanpur, India.

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Contents

- Introduction
- Statistical Process Control
- Design of Experiments
- Acceptance Sampling

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Introduction

- Quality has become one of the most important consumer decision factors in the selection among competing products and services.
- The Quality of a product can be decided by looking forward towards the following factors related to the product:
  1. Performance (will the product-do the intended job?)
  2. Reliability (how often does the product fail?)
  3. Durability (how long does the product last?)
  4. Serviceability (how easy is it to repair the product?)
  5. Aesthetics (what does the product look like?)
  6. Features (what does the product do?)
  7. Perceived Quality (what is the reputation of the company or its product?)
  8. Conformance to Standards (is the product made exactly as the designer intended?)

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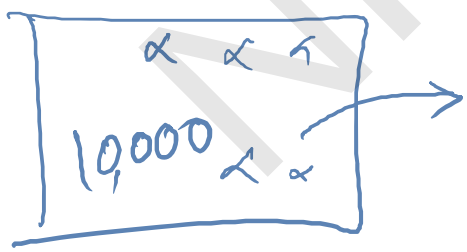
Statistical Methods For Quality Control and Improvement

In this method we will focus on three major areas:-

1. Statistical Process Control
2. Design of Experiments
3. Acceptance Sampling



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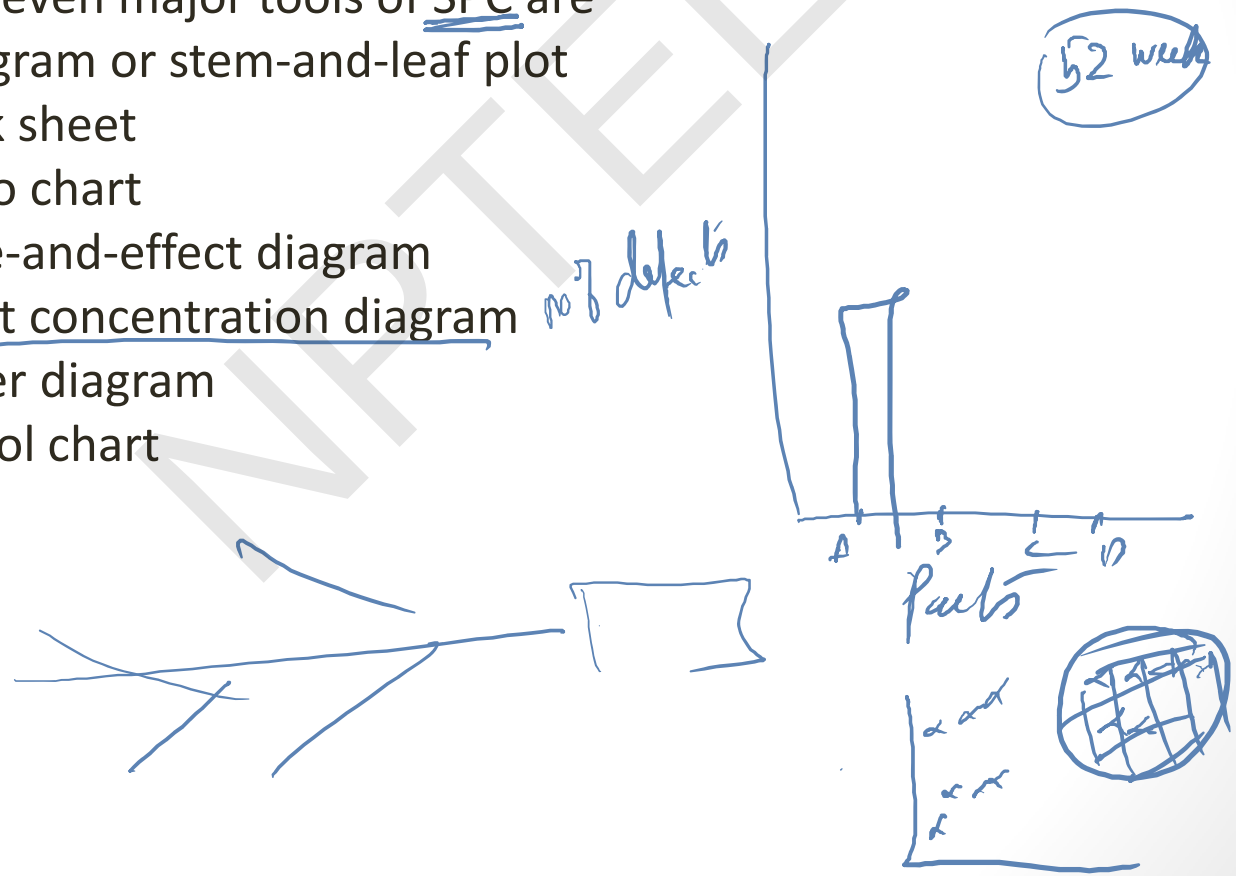
Statistical process control

- Statistical process control (SPC) is a powerful collection of problem-solving tools useful in achieving process stability and improving capability through the reduction of variability.
- The seven major tools of SPC are

1. Histogram or stem-and-leaf plot
2. Check sheet
3. Pareto chart
4. Cause-and-effect diagram
5. Defect concentration diagram
6. Scatter diagram
7. Control chart



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Statistical process control

Chance And Assignable Causes Of Quality Variation

- In any production process, a certain amount of inherent or natural-variability will always exist.
- This natural variability is the cumulative effect of many small, essentially unavoidable causes.
- Other type of variability may occasionally present in the output of a process.

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Statistical process control

Control Charts

- A control chart is one of the primary techniques of statistical process control.
- These type of charts plots the averages of measurements of a quality characteristic in samples taken from the process versus time (or the sample number).
- Control Chart is a very useful 'process monitoring technique'.
- Whenever unusual sources of variability are present, sample averages will plot outside the control limits.

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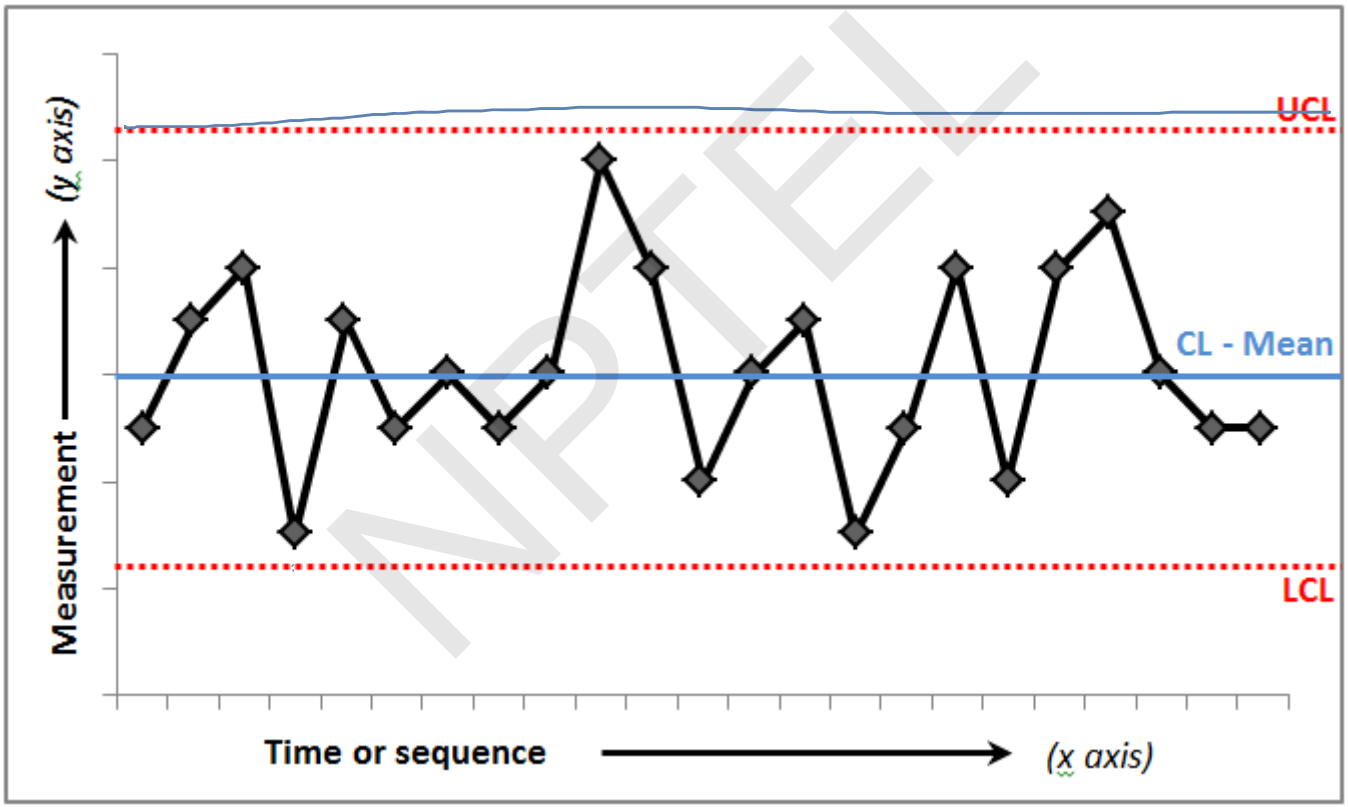
Statistical Basis of the Control Chart

Basic Principles



https://qi.elft.nhs.uk/wp-content/uploads/2014/08/typical-control-chart-1.png

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Statistical Basis of the Control Chart

Basic Principles

- The control chart is a graphical display of a quality characteristic that has been measured or computed from a sample versus the sample number or time.
- The chart contains a centre line that represents the average value of the quality characteristic corresponding to the in-control state.
- Two other horizontal lines, called the upper control limit (UCL) and the lower control limit (LCL), are also shown on the chart.
- These control limits are chosen so that if the process is in control, nearly all of the sample points will fall between them.
- As long as the points plot within the control limits, the process is assumed to be in control, and no action is necessary.

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Statistical Basis of the Control Chart

- We may give a general model for a control chart.
- Let w be a sample statistic that measures some quality characteristic of interest, and suppose that the mean of w is μ_w and the standard deviation of w is σ_w. Then the centre line, the upper control limit, and the lower control limit become

$$\text{UCL} = \mu_w + L\sigma_w$$
$$\text{Center line} = \mu_w$$
$$\text{LCL} = \mu_w - L\sigma_w$$

- where L is the "distance" of the control limits from the centre line, expressed in standard deviation units.

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Statistical Basis of the Control Chart

- The control chart is a device for describing in a precise manner exactly what is meant by statistical control; as such, it may be used in a variety of ways.
- In many applications, it is used for on-line process monitoring or surveillance.
- The most important use of a control chart is to improve the process.

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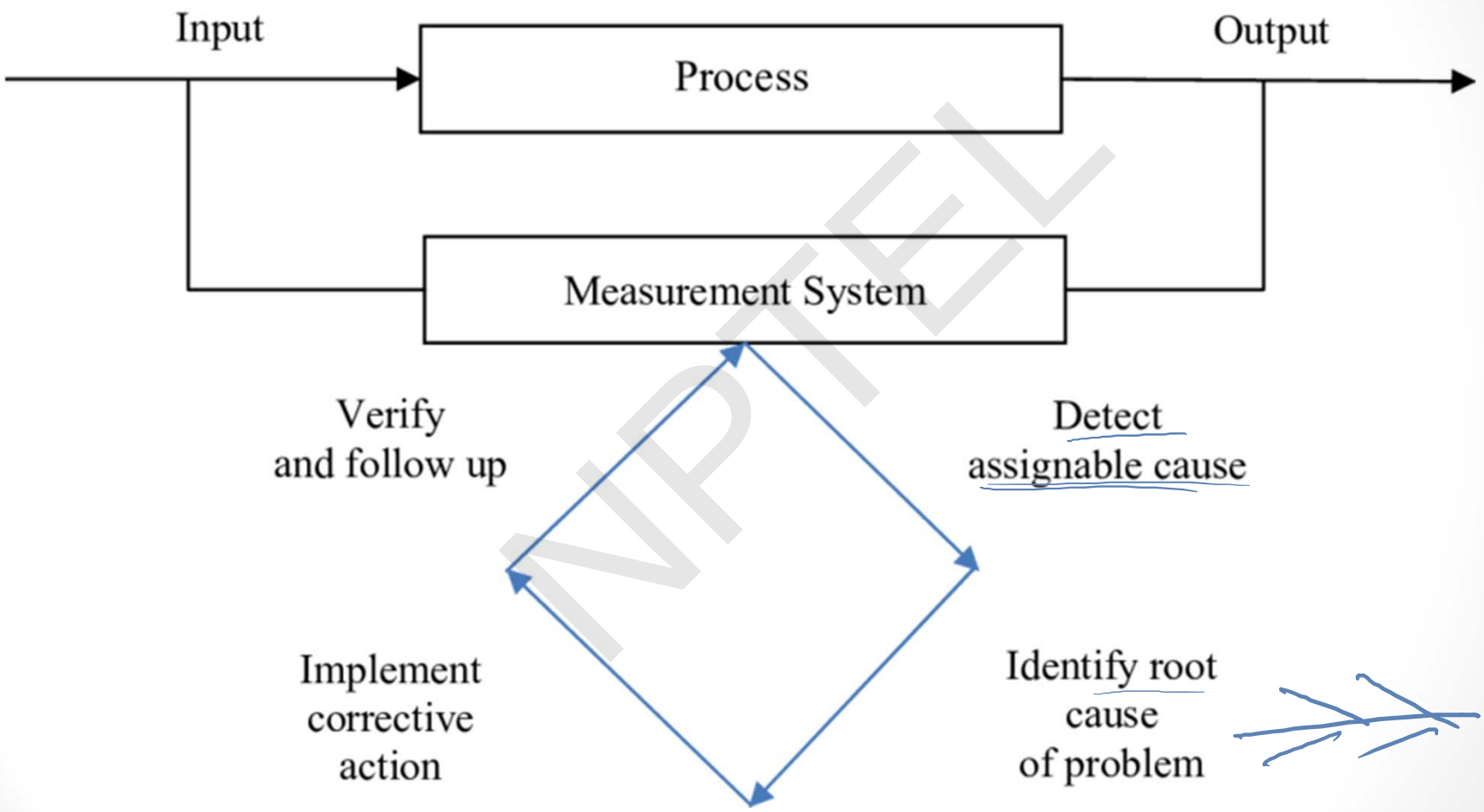
Statistical Basis of the Control Chart



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A lot, 2017

Statistical Basis of the Control Chart

- A very important part of the corrective action process associated with control chart usage is the out-of-control-action plan (OCAP).

- An OCAP is a flowchart or text-based description of the sequence of activities that must take place following the occurrence of an activating event.

Control charts are widely used for the following reasons:

1. Control charts are a proven technique for improving productivity
2. Control charts are effective in defect prevention
3. Control charts prevent unnecessary process adjustment
4. Control charts provide diagnostic information
5. Control charts provide information about process capability

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Statistical Basis of the Control Chart

Sample Size and Sampling Frequency

- In designing a control chart, we must specify both the sample size and the frequency of sampling.
- We can evaluate the decisions regarding sample size and sampling frequency is through the average run length (ARL) of the control chart.
- If the process observations are uncorrelated, then for any Shewhart control chart, the ARL can be calculated easily from

$$\boxed{\mathrm{ARL} = \frac{1}{p}}$$

- where p is the probability that any point exceeds the control limits.
- This equation can be used to evaluate the performance of the control chart.

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Statistical Basis of the Control Chart

Sample Size and Sampling Frequency

- It is also occasionally convenient to express the performance of the control chart in terms of its average time to signal (ATS).
- If samples are taken at fixed intervals of time that are h hours apart, then

$$\mathrm{ATS} = \mathrm{ARL}h$$

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Statistical Basis of the Control Chart

Phase I and Phase II of Control Chart Application

- Standard control chart usage involves phase I and phase II applications, with two different and distinct objectives.

- In phase I, a set of process data is gathered and analysed all at once in a retrospective analysis.
  - constructing trial control limits to determine if the process has been in control over the period of time during which the data were collected, and
  - to see if reliable control limits can be established to monitor future production.
- Control charts in phase I primarily assist operating personnel in bringing the process into a state of statistical control.

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Statistical Basis of the Control Chart

Phase I and Phase II of Control Chart Application

- Phase II begins after we have a "clean" set of process data gathered under stable conditions and representative of in-control process performance. In phase II, we use the control chart to monitor the process by comparing the sample statistic for each successive sample as it is drawn from the process to the control limits.

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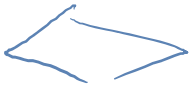
Statistical Basis of the Control Chart

Applications of Statistical Process Control and Quality Improvement Tools in Transactional and Service Businesses

- Flowcharts, operation process charts, and value stream mapping are particularly useful in developing process definition and process understanding.
- A flowchart is simply a chronological sequence of process steps or work flow. Sometimes flowcharting is called process mapping.
- Flowcharts or process maps must be constructed in sufficient detail to identify value-added versus non-value-added work activity in the process



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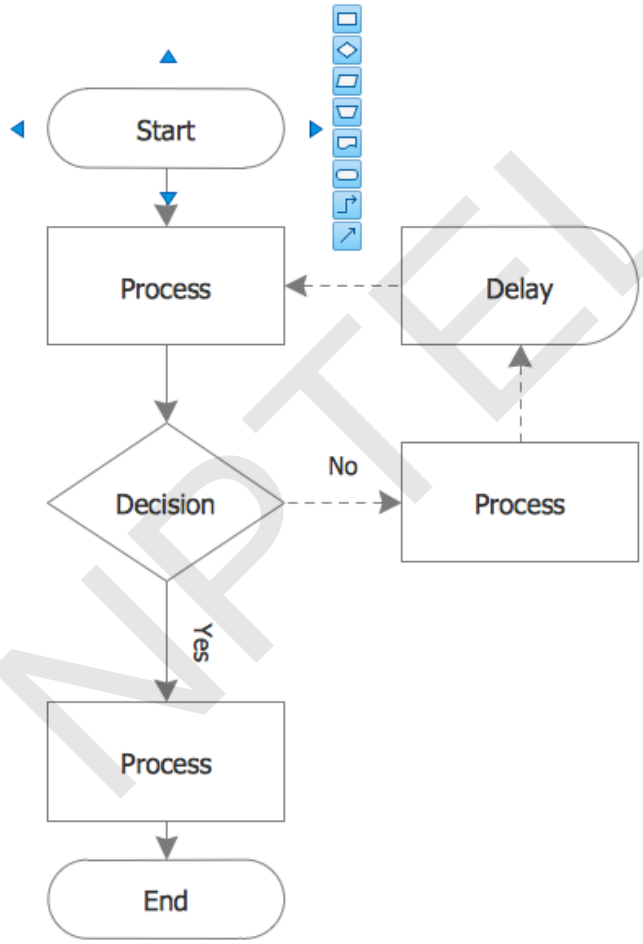
Statistical Basis of the Control Chart



http://www.conceptdraw.com/How-To-Guide/picture/process-flow-chart/Basic-Flowchart-template.png

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Statistical Methods For Quality Control and Improvement

Statistical Process Control

Control chart for variables are used to monitor characteristics that can be measured, e.g. length, weight, diameter, time, etc.

Control charts for attributes are used to monitor characteristics that have discrete values and can be counted, e.g. % defective, number of defects in a mobile phone, number of broken eggs in a box, etc.

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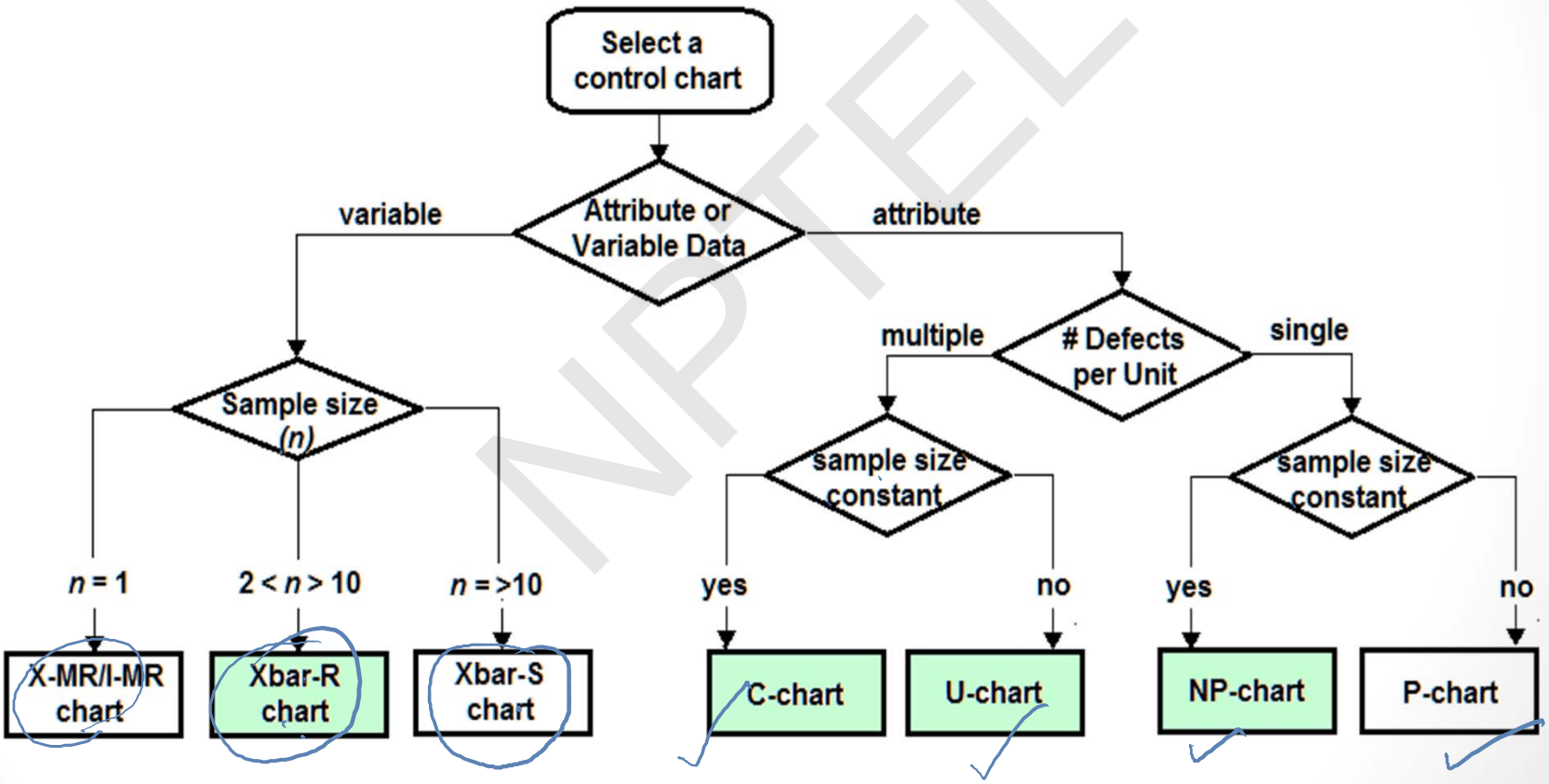
Statistical Methods For Quality Control and Improvement

Statistical Process Control



http://www.sixsigmatrainingfree.com/statistical-process-control-charts.html

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Control Charts for Variables

Xbar and R charts

- $X\ (UCL_x) = Xbar + A_2 * R$
- $X\ (LCL_x) = Xbar - A_2 * R$

  Where A_2 can be found from a standard table.

- $R\ (UCL_R) = D_4 * R$
- $R\ (LCL_R) = D_3 * R$

  Where D_4 and D_3 can be found from a standard table.

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Control Charts for Variables

Xbar and R charts

- To interpret the data we first look at the R chart.
- The R chart represents the variety in the data and if the variety is to great than there is no need to look at the X control chart.
- The chart is out of control if one or a combination of the following four examples occur:

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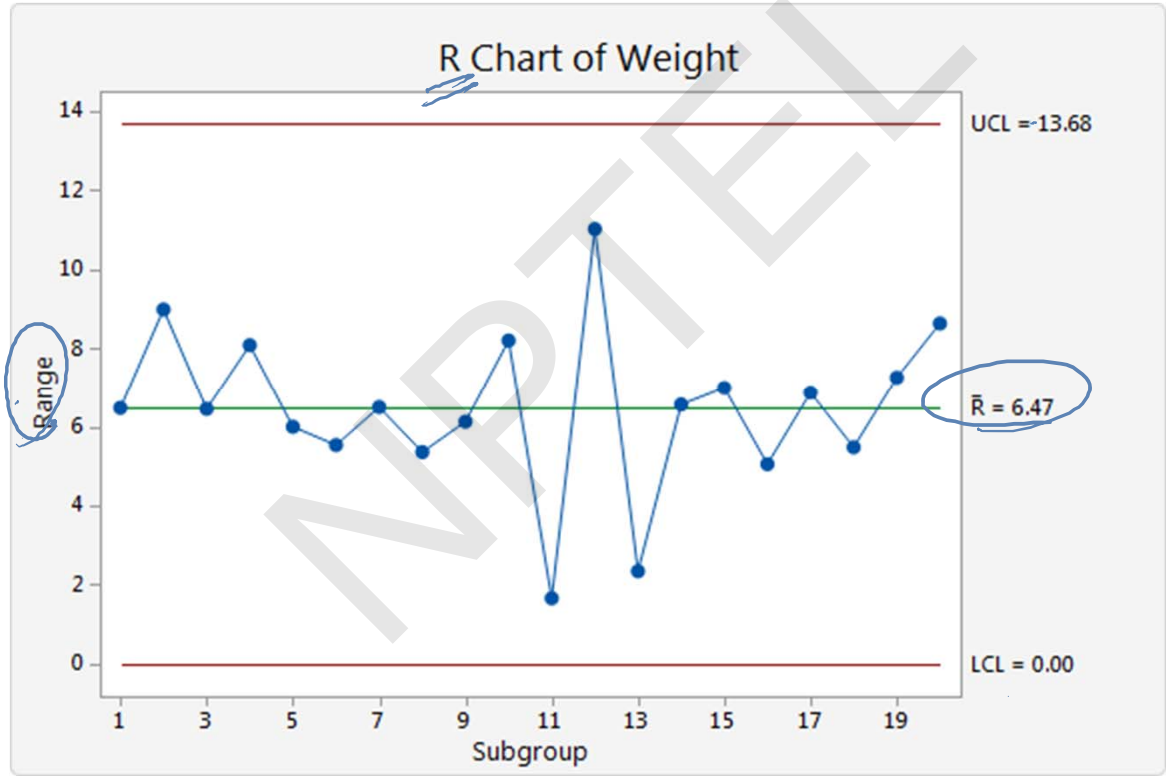
Statistical Methods For Quality Control and Improvement

Xbar and R charts



http://support.minitab.com/en-us/minitab-express/1/xrchart_key_results.xml_Graph_cmd1o2.png

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Control Charts for Attributes

p-charts:

- for quality characteristics that are discrete and involve yes/no or good/bad decisions
  - Percent of leaking caulking tubes in a box of 48
  - Percent of broken eggs in a carton

c-charts:

- for discrete defects when there can be more than one defect per unit
  - Number of flaws or stains in a carpet sample cut from a production run
  - Number of complaints per customer at a hotel

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Control Charts for Attributes

Attributes Control Charts:

- counts [c-chart]
- when sample size are not constant and/or are unknown, use counts charts [c-charts]

$$CL = \frac{\#\text{ of defects}}{\#\text{ of samples}}$$

$$UCL = \bar{c} + z\sqrt{\bar{c}}$$

$$LCL = \bar{c} - z\sqrt{\bar{c}}$$

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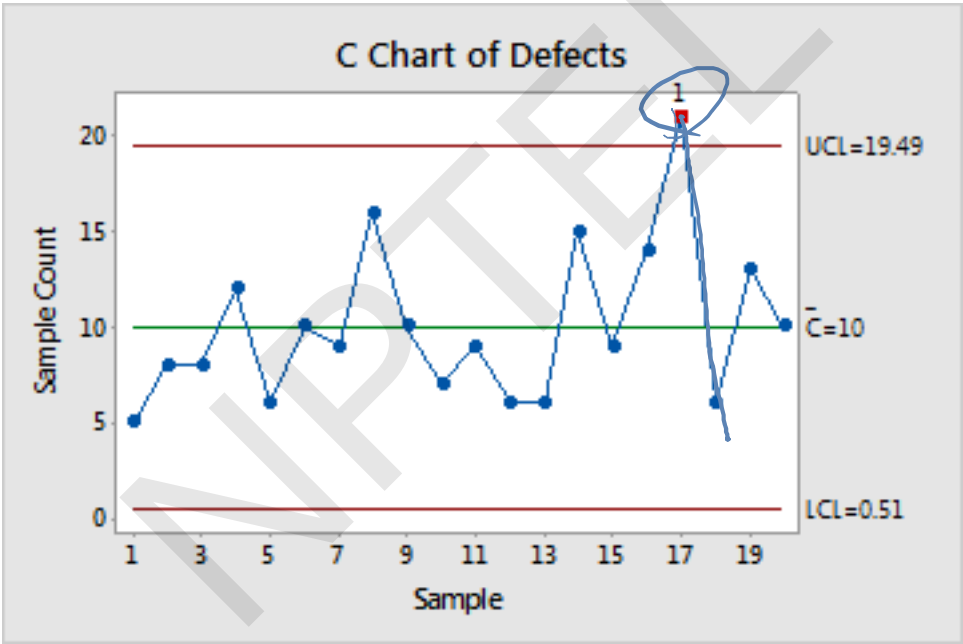
Statistical Methods For Quality Control and Improvement

Statistical Process Control



https://support.minitab.com/en-us/minitab/18/C_chart_def.png

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Control Charts for Attributes

Attributes Control Charts:

- proportion [p-charts]

$$UCL = \bar{p} + z\left(\sigma_p\right)$$

$$LCL = \bar{p} - z\left(\sigma_p\right)$$

$$\sigma_p = \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

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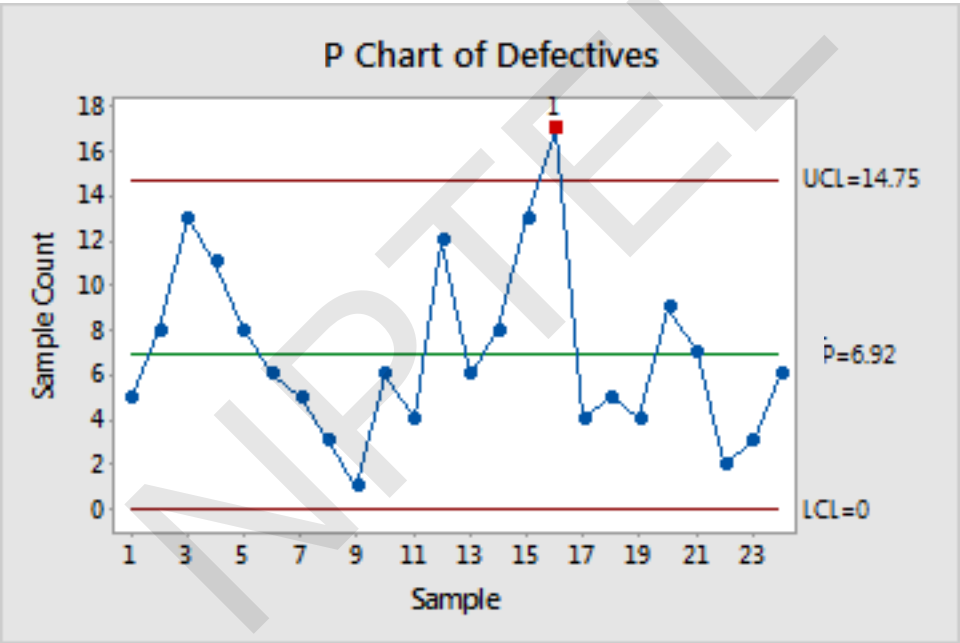
Statistical Methods For Quality Control and Improvement

Statistical Process Control



https://support.minitab.com/en-us/minitab/18/np_chart_of_defective_bulbs.png

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Control Charts for Nonconformities (Defects)

- A nonconforming item is a unit of product that does not satisfy one or more of the specifications for that product.
- Each specific point at which a specification is not satisfied results in a defect or nonconformity.
- Consequently, a nonconforming item will contain at least one nonconformity.
- As an example, suppose we are manufacturing personal computers. Each unit could have one or more very minor flaws in the cabinet finish, and since these flaws do not seriously affect the unit's functional operation, it could be classified as conforming.
- It is possible to develop control charts for either the total number of nonconformities in a unit or the average number of nonconformities per unit.

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Control Charts for Nonconformities (Defects)

Further Analysis of Nonconformities

- Defect or nonconformity data are always more informative than fraction nonconforming, because there will usually be several different types of nonconformities.
- By analysing the nonconformities by type, we can often gain considerable insight into their cause.
- This can be of considerable assistance in developing the out-of-control-action plans (OCAPs) that must accompany control charts.
- Another useful technique for further analysis of nonconformities is the cause-and effect diagram.

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Control Charts for Nonconformities (Defects)

The u Chart

- The second approach involves setting up a control chart based on the average number of nonconformities per inspection unit.
- If we find x total nonconformities in a sample of n inspection units, then the average number of nonconformities per inspection unit is

$$u = \frac{x}{n}$$

Note the *x* is a Poisson random variable.

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Control Charts for Nonconformities (Defects)

The u Chart

- Consequently, the parameters of the control chart for the average number of nonconformities per unit are as follows

$$\text{UCL} = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$$

$$\text{Center line} = \bar{u}$$

$$\text{LCL} = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$$

- Where $\bar{u}$ represents the observed average number of nonconformities per unit in a preliminary set of data.
- This per-unit chart often is called the control chart for nonconformities, or *u* chart.

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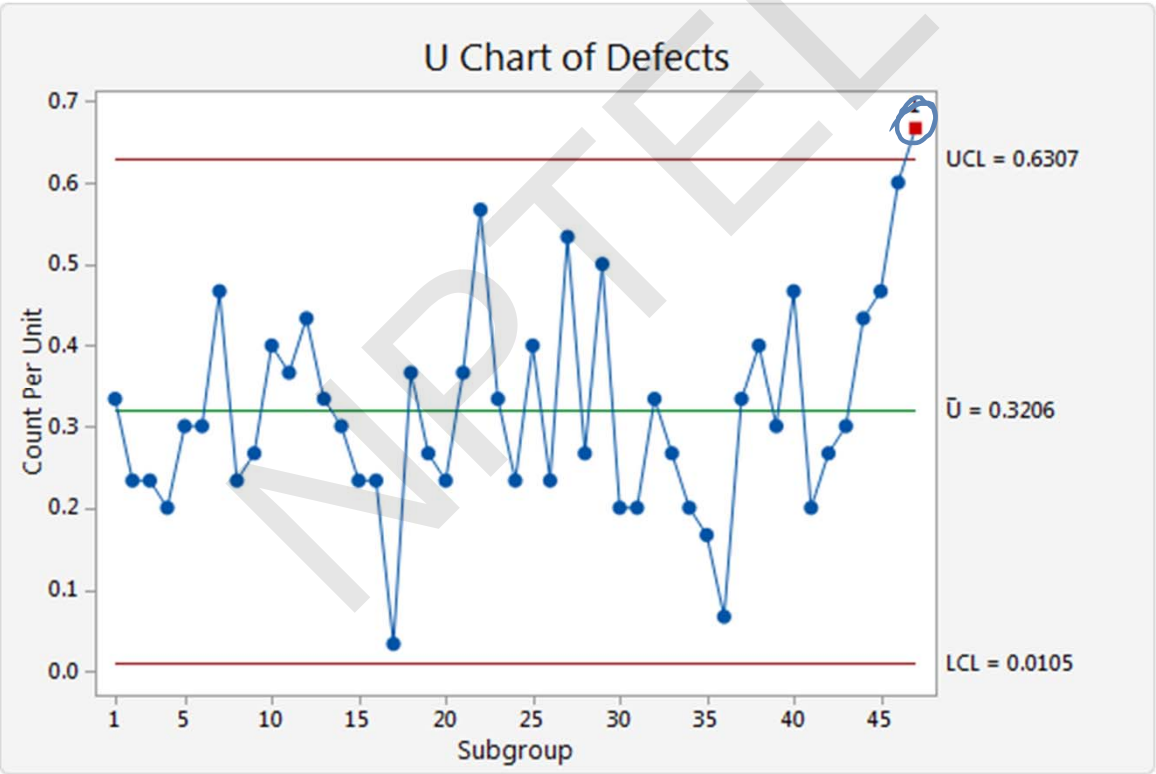
Statistical Methods For Quality Control and Improvement

Statistical Process Control



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Control Charts for Nonconformities (Defects)

Choice Between Attributes and Variables Control Charts

- Attributes control charts have the advantage that several quality characteristics can be considered jointly and the unit classified as nonconforming if it fails to meet the specification on any one characteristic.
- On the other hand, if the several quality characteristics are treated as variables, then each one must be measured, and either a separate and *R* chart must be maintained on each or some multivariate control technique that considers all the characteristics must simultaneously be employed.
- There is an obvious simplicity associated with the attributes chart in this case.
- Furthermore, expensive and time-consuming measurements may sometimes be avoided by attributes inspection.

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Control Charts for Nonconformities (Defects)

Choice Between Attributes and Variables Control Charts

- Variables control charts, in contrast, provide much more useful information about process performance than do attributes control charts.
- Specific information about the process mean and variability is obtained directly. In addition, when points plot out of control on variables control charts, usually much more information is provided relative to the potential cause of that out-of-control signal.
- For a process capability study, variables control charts are almost always preferable to attributes control charts.
- The exceptions to this are studies relative to nonconformities produced by machines or operators in which there are a very limited number of sources of nonconformities, or studies directly concerned with process yields and fallouts.

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Control Charts for Nonconformities (Defects)

Guidelines for Implementing Control Charts

1. Determining which process characteristics to control
2. Determining where the charts should be implemented in the process
3. Choosing the proper type of control charts
4. Taking actions to improve processes as the result of SPC/control chart analysis
5. Selecting data-collection systems and computer software

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Statistical Methods For Quality Control and Improvement

Design of Experiments

- It is extremely helpful in discovering the key variables influencing the quality characteristics of interests in the process.
- It is an approach to systematically varying the controllable input factors in the process the effect these factors have on the output product parameters.
- The major type of designed experiment is the factorial design.
- In the factorial design, factors are varied together in such a way that all possible combinations of factor levels are tested.

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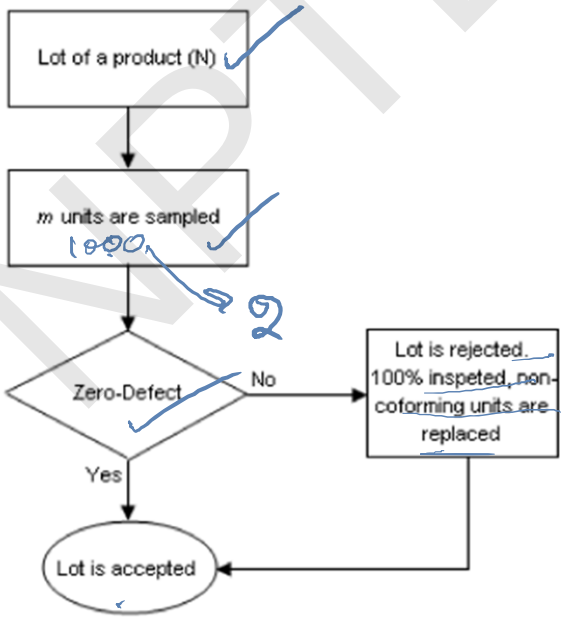
Statistical Methods For Quality Control and Improvement

Acceptance Sampling

- Acceptance Sampling is connected with inspection and testing of product.



http://www.scielo.br/img/revistas/pope/v25n1/24249f1.gif

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Task for students

Note down an event which you do regularly

- Wake up —
- Leave to office —
- Coming back home —
- Sleep —

At least 10 days

$x_1, x_2, x_3, x_4, x_5 \ldots x_{10} = \mu$

$= \sigma$

$\mu \pm 2\sigma \quad \mu \pm 3\sigma$

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Thanks

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Quality Assurance

NPTEL

Dr. Janakarajan Ramkumar

Professor

Department of Mechanical & Design Program

IIT Kanpur, India.

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- Process Capability Analysis Using a Control Chart
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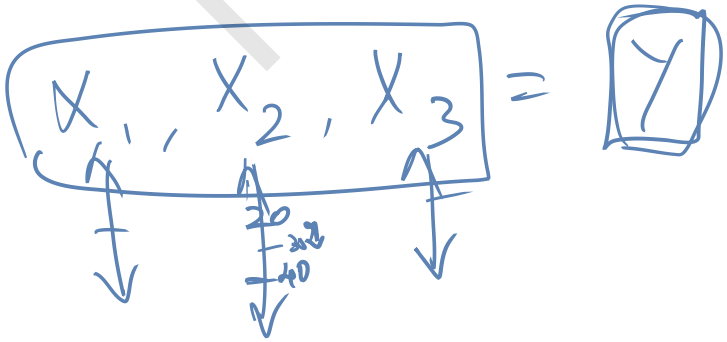
- Engineering Process Control (EPC) and SPC
- Process Monitoring and Process Regulation
- Process Control by Feedback Adjustment
- Combining SPC and EPC

NPTEL

Introduction

- Statistical techniques can be helpful throughout the product cycle, including development activities prior to manufacturing:
  - in quantifying process variability,
  - in analysing this variability relative to product requirements or specifications, and
  - in assisting development and manufacturing in eliminating or greatly reducing this variability.

- This general activity is called process capability analysis.



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Process Capability

- **Process capability** refers to the uniformity of the process.
- The variability of critical-to-quality characteristics in the process is a measure of the uniformity of output.

There are two ways to think of this variability:

1. The natural or inherent variability in a critical-to-quality characteristic at a specified time—that is, "instantaneous" variability
2. The variability in a critical-to-quality characteristic over time

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Process Capability Ratios

VSL or VCL

Use and Interpretation of C_p

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$$

$$C_{pu} = \frac{\text{USL} - \mu}{3\sigma} \quad \text{(upper specification only)}$$

$$C_{pl} = \frac{\mu - \text{LSL}}{3\sigma} \quad \text{(lower specification only)}$$

$$\hat{C}_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}}$$

USL

μ

LSL

- where USL and LSL are the upper and lower specification limits, respectively.

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Process Capability Ratios

Process Capability Ratio for an Off-Centre Process

- The process capability ratio C_p does not take into account where the process mean is located relative to the specifications.
- C_p simply measures the spread of the specifications relative to the Six Sigma spread in the process.

- This situation may be more accurately reflected by defining a new process capability ratio (PCR)—C_{pk}—that takes process centering into account.

$$C_{pk} = \min\left(C_{pu}, C_{pl}\right)$$

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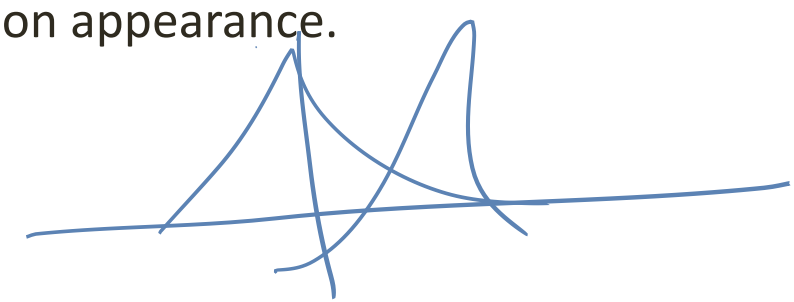
Process Capability Ratios

Normality and the Process Capability Ratio

- An important assumption underlying our discussion of process capability and the ratios C_p and C_{pk} is that their usual interpretation is based on a normal distribution of process output.

- If the underlying distribution is non-normal, then, the statements about expected process fallout attributed to a particular value of C_p or C_{pk} may be in error.

- One approach to dealing with this situation is to transform the data so that in the new, transformed metric the data have a normal distribution appearance.



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Process Capability Ratios

Normality and the Process Capability Ratio

- Other approaches have been considered in dealing with non-normal data.
- There have been various attempts to extend the definitions of the standard capability indices to the case of non-normal distributions.
- Luceño (1996) introduced the index C_{pc}, defined as

$$C_{pc} = \frac{\text{USL} - \text{LSL}}{6\sqrt{\frac{\pi}{2}}E|X - T|}$$

E = ?

- where the process target value

$$T = \tfrac{1}{2}(\text{USL} + \text{LSL})$$

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Process Capability Analysis Using a Control Chart

- Histograms, probability plots, and process capability ratios summarize the performance of the process.
- They do not necessarily display the potential capability of the process because they do not:
  - address the issue of statistical control, or
  - show systematic patterns in process output that, if eliminated, would reduce the variability in the quality characteristic.

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Process Capability Analysis Using a Control Chart

- Control charts are very effective in this regard.
  - The control chart should be regarded as the primary technique of process capability analysis.
  - Both attributes and variables control charts can be used in process capability analysis.
  - The and *R* charts should be used whenever possible, because of the greater power and better information they provide relative to attributes charts.
  - However, both *p* charts and *c* (or *u*) charts are useful in analysing process capability.

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Process Capability Analysis Using DoE

- A designed experiment is a systematic approach to varying the input controllable variables in the process and analysing the effects of these process variables on the output.

- Designed experiments are also useful in discovering:
  - which set of process variables is influential on the output, and
  - at what levels these variables should be held to optimize process performance.

- One of the major uses of designed experiments is in isolating and estimating the sources of variability in a process.

$x_1, x_2, x_3 = Y$

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Engineering Process Control (EPC) and SPC

- There are two statistically based approaches for addressing this problem:
  - The first of these is statistical process monitoring by control charts, or statistical process control (SPC).
  - The focus of SPC is on identifying assignable causes so that they can be removed, thereby leading to permanent process improvement or reduction in variability.
  - The second approach is based on adjusting the process using information about its current level or deviation from a desired target.
  - This approach is often called feedback adjustment, and it is a form of Engineering Process Control or EPC.
- Feedback adjustment regulates the process to account for sources of variability that cannot be removed by the SPC approach.

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Engineering Process Control (EPC) and SPC

- **Engineering Process Control (EPC)** is a process compensation or regulation schemes are widely known as stochastic control, or feedback or feedforward control, depending on the nature of the adjustments.

- This approach is based on process compensation and regulation, in which some manipulatable process variable is adjusted with the objective of keeping the process output on target (or equivalently, minimizing the variability of the output around this target).

- SPC has had a long history of successful use in discrete parts manufacturing.
- EPC is used in continuous processes, such as those found in the chemical and process industries to reduce variability.

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Process Control by Feedback Adjustment

A Simple Adjustment Scheme: Integral Control

- In this we consider a simple situation involving a process in which feedback adjustment is appropriate and highly effective.
- The process output characteristic of interest at time period t is y_t, and we wish to keep y_t as close as possible to a target T. This process has a manipulatable
- variable x, and a change in x will produce all of its effect on y within one period—that is,

$$y_{t+1} - T = gx_t$$

- where g is a constant usually called the **process gain.** The gain is like a regression coefficient, in that it relates the magnitude of a change in x_t to a change in y_t.
- Now, if no adjustment is made, the process drifts away from the target according to

$$y_{t+1} - T = N_{t+1}$$

- where $N_t + 1$ is a **disturbance.**

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Process Control by Feedback Adjustment

The Adjustment Chart

- When EPC or feedback adjustment is implemented in this manner, it is often called automatic process control (APC).
- In many processes, feedback adjustments can be made manually.
- Operating personnel routinely observe the current output deviation from target, compute the amount of adjustment to apply, and then bring x_t in the following equation to its new set point.

$$x_t - x_{t-1} = -\frac{\lambda}{g}(y_t - T) = -\frac{\lambda}{g}e_t$$

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Combining SPC and EPC

- There is considerable confusion about process adjustment versus process monitoring.
- Process adjustment or regulation has an important role in reduction of variability.
- There are many processes where some type of feedback-control scheme would be preferable to a control chart.
- In processes where feedback control is used there may be substantial improvement if control charts are also used for statistical process monitoring (as opposed to control; the control actions are based on the engineering scheme).

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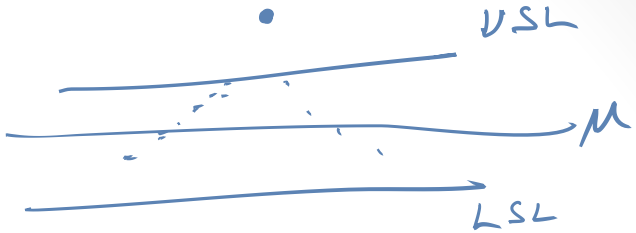
Combining SPC and EPC



- The control chart should be applied either
  - to the control error (the difference between the controlled variable and the target) or
  - to the sequence of adjustments to the manipulated variable. Combinations of these two basic approaches are also possible.

- Points that lie outside the control limits on these charts would identify periods during which the control errors are large or during which large changes to the manipulated variable are being made.

- These periods would likely be good opportunities to search for assignable causes.

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Task for students

daily	→ get up in the morning	going to bed
1	→ 6:15	9:00 pm
2	→ 7:20	11:30 pm
3	→ 5:30	2:00 am
4		
5		
6		
7		
8		
9		
10		

$\mu \Rightarrow$

$\sigma \Rightarrow$

$C_p =$; $C_{pk} =$

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Thanks

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